Fast Fourier Transform

Polynomials

$$A(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{n-1} x^{n-1}$$

$$= \sum_{k=0}^{n-1} a_k x^k$$

$$= \langle a_0, a_1, a_2, \ldots, a_{n-1} \rangle$$

eg $n=4$

$5 + 3x - 2x^2 + 10x^3$

$\langle 5, 3, -2, 10 \rangle$

operations

① evaluation

$A(x)$ for some given $x_0 \rightarrow A(x_0)$

runtime : # additions, multiplications

$\Theta(n^2)$ naive

$\Theta(n)$ smart

② addition

given $A(x)$, $B(x)$

want $C(x) = A(x) + B(x) \quad \forall x$

$c_k = a_k + b_k \quad \Theta(n)$

③ multiplication

given $A(x), B(x)$

want $C(x) = A(x) \cdot B(x) \ \forall x$

$$(a_0 + a_1 x + a_2 x^2 + \dots) \cdot (b_0 + b_1 x + b_2 x^2 + \dots)$$

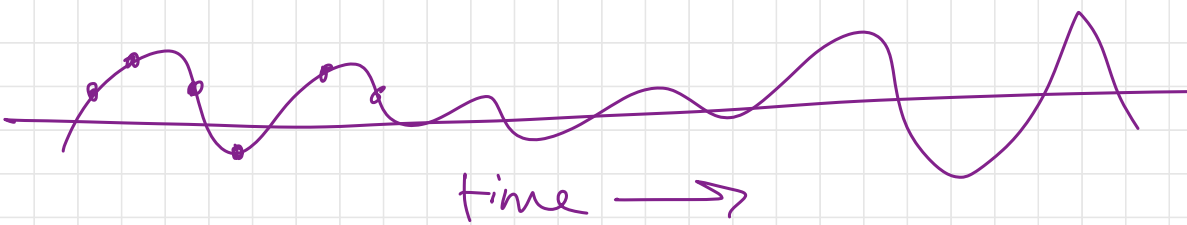
$$= a_0 b_0 + a_0 b_1 x + b_0 a_1 x + \dots$$

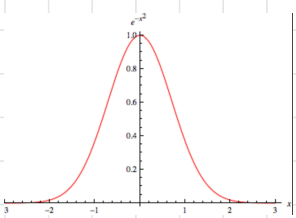
$$\boxed{c_k} = \boxed{\sum_{j=0}^{k} \underbrace{a_j}_{x^j} \underbrace{b_{k-j}}_{x^{k-j}}}$$

$$x^j x^{(k-j)} = x^k$$

$\Theta(n^2)$ – $\times$ mults per coefficient

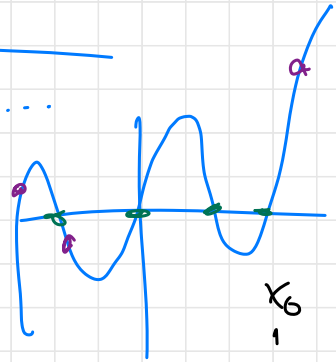
Vector convolution





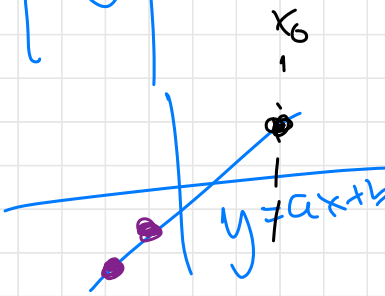
Representations of polynomials

(A) coeffs : $a_0 + a_1 x + a_2 x^2 + \cdots$



(B) roots $r_0, r_1, \ldots, r_{n-1}$

$c(x - r_0)(x - r_1) \cdots (x - r_{n-1})$



(C) samples

(x_k, y_k) for $k = 0, 1, \ldots, n-1$

$x_0 = 0$ $x_1 = 1$

s.t. $A(x_k) = y_k$

x_k s distinct

	(A) coeffs	(B) roots	(C) samples
(1) evaluation	$O(n)$	$O(n)$	$O(n^2)$
(2) addition	$O(n)$	∞	$O(n)$
(3) multiplication	$O(n^2)$	$O(n)$	$O(n)$

$O(n \log n)$

sampling Points $\rightarrow$ V matrix vandermonde

$$\begin{pmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & & \ddots & \\ 1 & x_{n-1} & & & x_{n-1}^{n-1} \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{n-1} \end{pmatrix} = \begin{pmatrix} y_0 \\ y_1 \\ y_2 \\ \vdots \\ y_{n-1} \end{pmatrix}$$

coeffs $\rightarrow$ samples $\quad V \cdot A$, $O(n^2)$

Samples $\rightarrow$ coeffs gaussian elim. $O(n^3)$

if we had V^{-1} $\quad V^{-1}Y$, $O(n^2)$

Divide + Conquer

goal: $A(x)$ for $x \in X$

1. Divide into even coeffs + odd coeffs

$A_{even} = \langle a_0, a_2, \ldots \rangle$

$A_{odd} = \langle a_1, a_3, \ldots \rangle$

2 conquer

3 combine